

This project is a design and implementation of an FPGA-based multi-effects system for the electric guitar, on the Zedboard.

It is designed to work with a guitar input in the Line-in port, and an amplifier in the Line-out.

The effects we created are: Distortion/OD, Octavolo (an experimental effect which combines Octaver and Tremolo),

Tremolo, and Delay, where it's possible to activate each effect individually, or concatenate several effects in series, to produce interesting combinations.

We also implemented a user interface, based on the Zedboard's buttons, switches and LEDs, for real-time user control of the effects.

The ip_repo folder contains all of the packaged IPs - suitable for IP import.

To run the project:

1. Extract both archives to a single folder named Meffects_constants_testing_3, which after the extract shall include 7 folders and one .xpr file.
2. Open the .xpr file from the Meffects_constants_testing_3 folder, it already includes a bitstream file.
3. Set the IP repository folder to be ip_repo.
4. Launch SDK
5. Program FPGA (Blue LED on Zedboard should light up)
6. Right-click on the Audio project -> Run as -> Launch On Hardware (GDB) - the leftmost red LED on Zedboard should light up.
7. Connect an audio source to the Line-in port, and amplifier/speakers to the Line-out or Headphones outputs.

8. Listen to the output, and use the slide-switches and buttons to change effects.

The 4 leftmost switches of the Zedboard activate each of the 4 different effects.

The 4 rightmost switches of the Zedboard choose the internal configuration of each effect.

'1' means the switch is up, and '0' is down.

For example - 1000 + 1000 means the first effect switch is up (Distortion), and inside, the first configuration is chosen (Weak OD1).

1000 (Distortion) + 1000 (Weak OD1) / 0100 (Strong OD) / 0010 (Weak OD2) / 0001 (Harsh Distortion)

0100 (Octavolo) + 1000 (FIR, 1up) / 1100 (FIR2, 1up) / 1110 (FIR3, 1up) / 1111 (IIR3, 1up) / 0111 (IIR2, 1up) / 0011 (IIR1, 1up) / 0100 (FIR, 2up) / 0010 (FIR, 1down) / 0001 (FIR, robot)

0010 (Tremolo) + 1000 (1.6Hz) / 0100 (3.5Hz) / 0010 (6.35Hz) / 0001 (0.8Hz)

0001 (Delay) + 1000 (IIR1) / 1100 (IIR2,short) / 1110 (IIR3, slight delay-reverb) / 0100 (IIR4, very long) / 0010 (FIR, single rep. long)